

STAGEPILOT (STAGEFLOW)

Project Progress Report & GPT Context Handover | Date: Sep 19, 2026

1. Project Architecture & Overview

StagePilot (StageFlow) is a real-time multi-track stage management and live teleprompter application. It empowers event organizers to control schedules, manage speaker delays across multiple stages, and provide anchors with a synchronized, high-contrast teleprompter featuring real-time speech tracking and AI assistance.

Component	Technology & Setup
Repository	https://github.com/rudra0279/StageFlow.git
Frontend App	React + Vite (Port 5173 / 5175), Web Audio API, Web Speech API
Backend Server	Node.js, Express, Socket.IO (Port 5001 - port 5000 fallback handled)
Database	MongoDB (with In-Memory Server fallback & auto-seeding for local dev)
AI Provider	Google Gemini AI API (Script generation & Stage Co-Pilot)

2. Stage Implementation Matrix

Stage / Feature	Description	Status
Stage 1: Anchor Teleprompter	High-contrast UI, mirror mode (flip horizontally), auto-scroll, speed/font controls, pronunciation guide, urgent flash banner, Web Audio chime, Browser STT Speech Follower.	COMPLETED
Stage 2: Organizer War Room	Real-time control dashboard, Cascading Delay Engine across sessions, AI Stage Co-Pilot (Gemini API), Socket.IO live sync.	COMPLETED
Stage 3: Multi-Track System	Track A (Main Stage), Track B (Workshop Hall), Track C (Networking) support, track-isolated delay engine, track selector in Anchor view, 13/13 test suite passing.	COMPLETED
Infrastructure & Fixes	Fixed missing helmet module, resolved macOS Port 5000 AirPlay conflict by moving server to Port 5001, updated Vite proxy config, pushed all commits to remote main.	RESOLVED & PUSHED

3. Operational Credentials & Endpoint Verification

- **Backend API Endpoint:** <http://localhost:5001/api> (Health Check: <http://localhost:5001/api/health>)
- **Frontend Client App:** <http://localhost:5173> or <http://localhost:5175>
- **Demo Account (Organizer):** organizer@stagepilot.io | Password: password123
- **Demo Account (Anchor):** anchor@stagepilot.io | Password: password123

4. Ready-to-Use Prompt Template for GPT

Hi GPT! I am continuing work on StagePilot (StageFlow), a real-time multi-track stage management and teleprompter application.

Current Architecture & Completed Features:

1. Tech Stack: Node.js, Express, Socket.IO, MongoDB, React, Vite, Web Speech API, Gemini AI.
2. Teleprompter: High-contrast stage view, mirror mode, auto-scroll, Web Speech STT follower, audio announcements.
3. Organizer War Room: Live control dashboard, Cascading Delay Engine, AI Stage Co-Pilot.
4. Multi-Track Engine: Track A, Track B, Track C support with track-isolated delay management.
5. Deployment & Remote: Server on Port 5001, Client proxy configured, all code pushed to GitHub (<https://github.com/rudra0279/StageFlow.git>).

I would like to implement the following next task/feature:

[INSERT YOUR NEXT FEATURE HERE - e.g. 'Add PDF export for stage schedules', 'Build presenter timer HUD', or 'Add speaker analytics']

Please analyze my architecture and provide a step-by-step technical implementation plan.